FISH FEEDING AND WATER QUALITY MANAGEMENT

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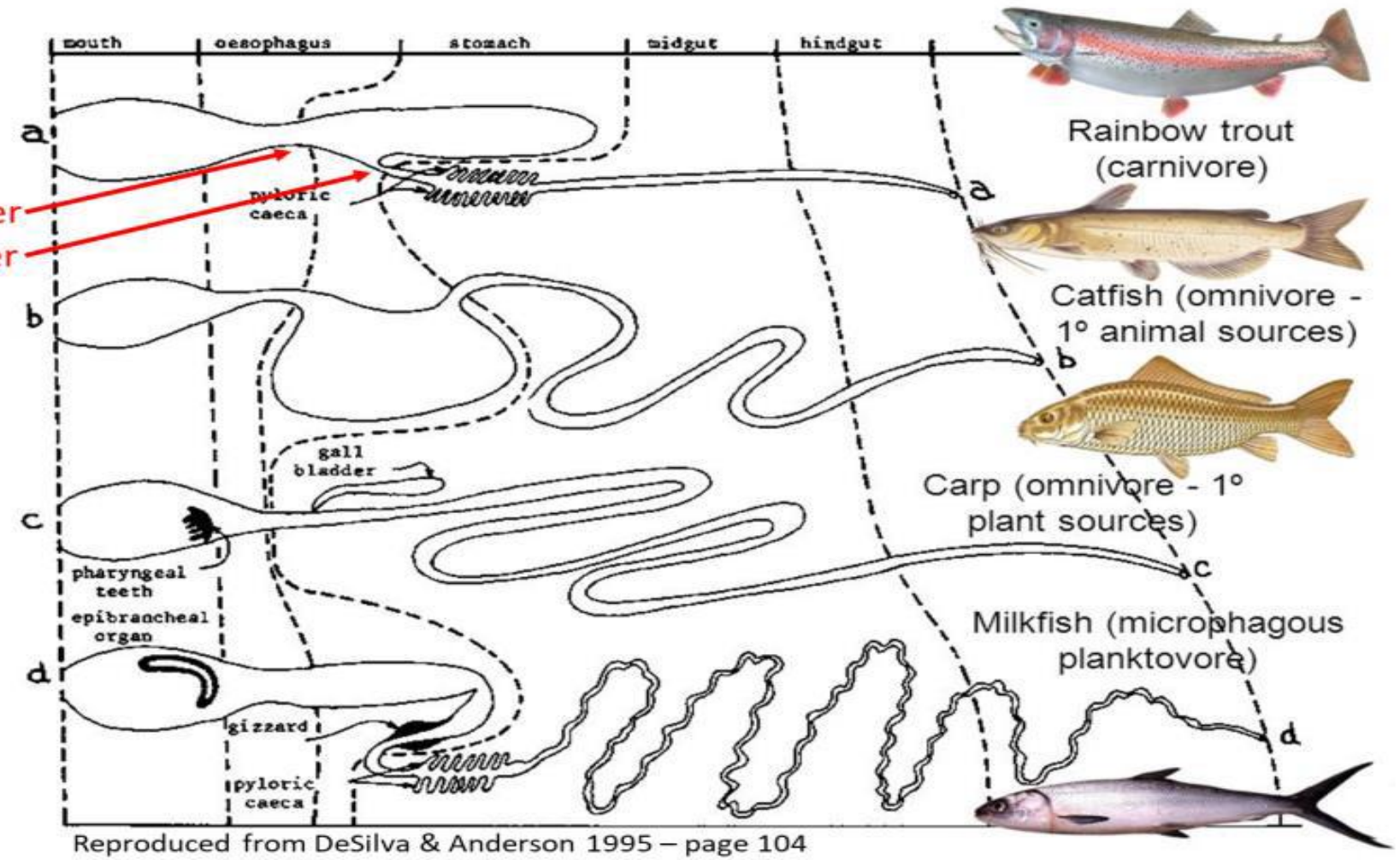
Fish feeding

To appropriate provide the much needed feed to the fish, there is need to understand the type of fish based on the following key things;

1. If there are classified has (Herbivorous Omnivorous and Carnivorous)

Fish feeding

Oesophageal sphincter
Pyloric sphincter



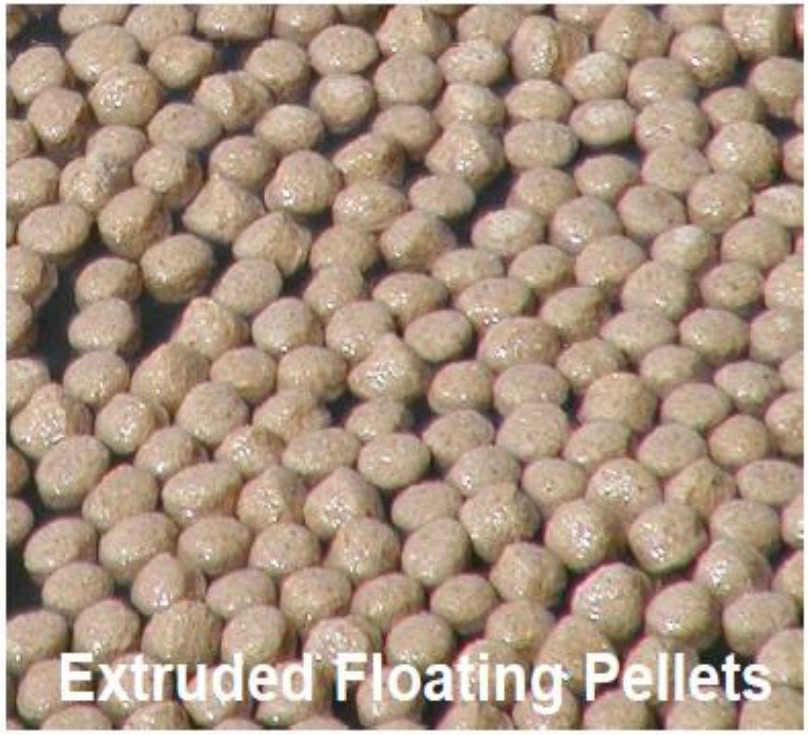
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Fish feeding

2. If there are bottom feeders or surface feeders

- Floating or sinking feed must be provided based on the nature of the species





Fish feeding

3.Type of the mouth

Terminal mouth – This type of a mouth is located at the end of the head of the fish (Tuna fish)



Fish feeding

4.Type of the mouth

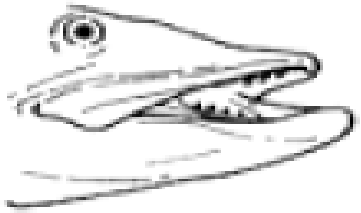
Sub-terminal – The mouth is on the lower side of the head



Fish feeding

4.Type of the mouth

Up-pointing – The fish has a longer lower jaw than the upper jaw. The mouth opening is toward the upper part of the head of the fish



Fish feeding

4. Life stages of fish

- The size and quantity of feed and feeding rates will be determined by the life stage of the fish.

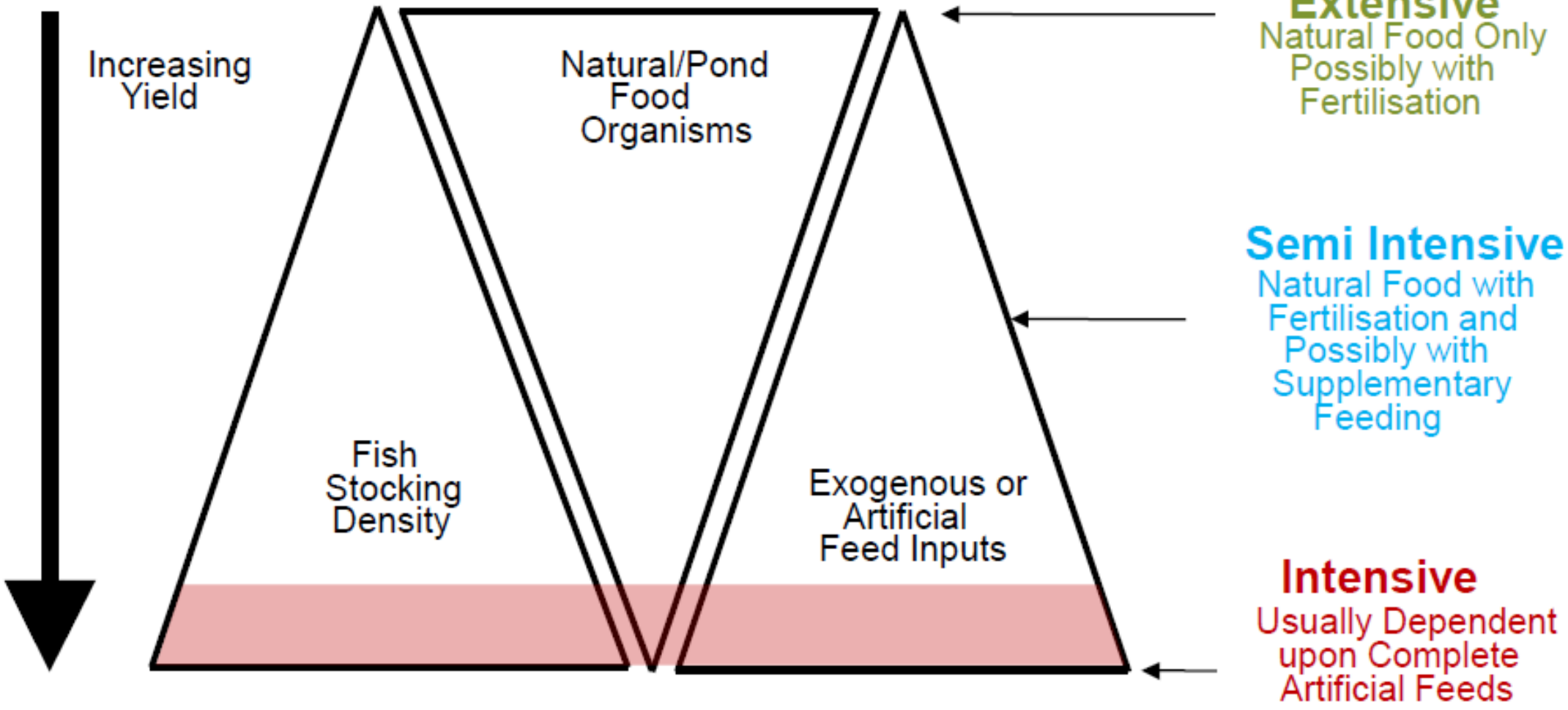
Feeding and levels of intensification in aquaculture

1.**Extensive aquaculture:** Natural food as only source; sometimes it may also involve fertilisation (to enhance production of aquatic organisms)

2. **Semi-intensive aquaculture:** Natural food with fertilisation and, when yield increases, supplemental feeding (incomplete feeds) is used

3. **Intensive aquaculture:** Use of nutritionally balanced diets (‘complete diets’) and minor or no inputs through natural food

Feeding and levels of intensification in aquaculture



TYPES OF FEED

1. Natural food (Produced in-situ in the culture facility)

2. Supplementary feed (Nutritionally complements natural food)

- Agricultural by-products e.g., rice bran, broken rice, and waste vegetables
- Domestic waste food e.g., from the household or restaurants

TYPES OF FEED

3.Complete feed





METHODS OF FEEDING

1. Manual method

- In this method of feeding the fish, an individual throws the feed into the facility/pond.
- It is advisable to feed from the same point and if possible at the same time.



METHODS OF FEEDING

1. Manual method



METHODS OF FEEDING

2. Automatic feeders

- This is where feeders are programmed to provide feed at specific times and quantities or can feed according to the detected demand.

Discussion

What are the advantages and disadvantages of manual and automatic feeding methods?

WATER QUALITY MANAGEMENT IN AQUACULTURE

1. Dissolved Oxygen

- What is the impact of photosynthesis and respiration of phytoplankton's on dissolved oxygen?
- Therefore, photosynthesis by phytoplankton is the primary source of dissolved oxygen in a fish culture system.
- Fish culturists are often concerned with the rate at which dissolved oxygen is removed from the water.
- The primary losses of dissolved oxygen include respiration by the plankton, respiration by fishes, respiration by benthic organisms.

WATER QUALITY MANAGEMENT IN AQUACULTURE

1. Dissolved Oxygen

TABLE 1.4. *Ranges of expected gains and losses of dissolved oxygen caused by different processes in fish ponds, for ponds of 1.0 to 1.5 metres average depth*

Process		Range mg/litre
Gains		
Photosynthesis by phytoplankton	..	5 to 20
Diffusion	..	1 to 5
Losses		
Plankton respiration	..	5 to 15
Fish respiration	..	2 to 6
Respiration by organisms in mud	..	1 to 3
Diffusion	..	1 to 5

WATER QUALITY MANAGEMENT IN AQUACULTURE

1. Dissolved Oxygen

- Concentrations of dissolved oxygen are lowest in the early morning just after sunrise, increase during daylight hours to a maximum in late afternoon and decrease again during the night.
- The magnitude of fluctuation is greatest in ponds with heavy plankton blooms and least in ponds with low plankton abundance.

WATER QUALITY MANAGEMENT IN AQUACULTURE

1. Dissolved Oxygen

- Dissolved oxygen can be increased by aerating the water which increases the diffusion of oxygen in water.
- You can also continuously pump water into the pond or drain 1/3 of the water and replace with new water

WATER QUALITY MANAGEMENT IN AQUACULTURE

2. pH (measure of acidity or alkalinity)

- chemical parameter to consider because it affects the metabolism and other physiological processes of culture organisms.
- A certain range of pH (pH 6.8 – 8.7) should be maintained for acceptable growth and production.

WATER QUALITY MANAGEMENT IN AQUACULTURE

2. pH (measure of acidity or alkalinity)

- The rate of CO2 consumption depends on phytoplankton density and is greatly influenced by the concentration of carbon dioxide
- Phytoplankton and other aquatic vegetation remove carbon dioxide from the water during photosynthesis, so the pH of a body of water rises during the day and decreases during the night

WATER QUALITY MANAGEMENT IN AQUACULTURE

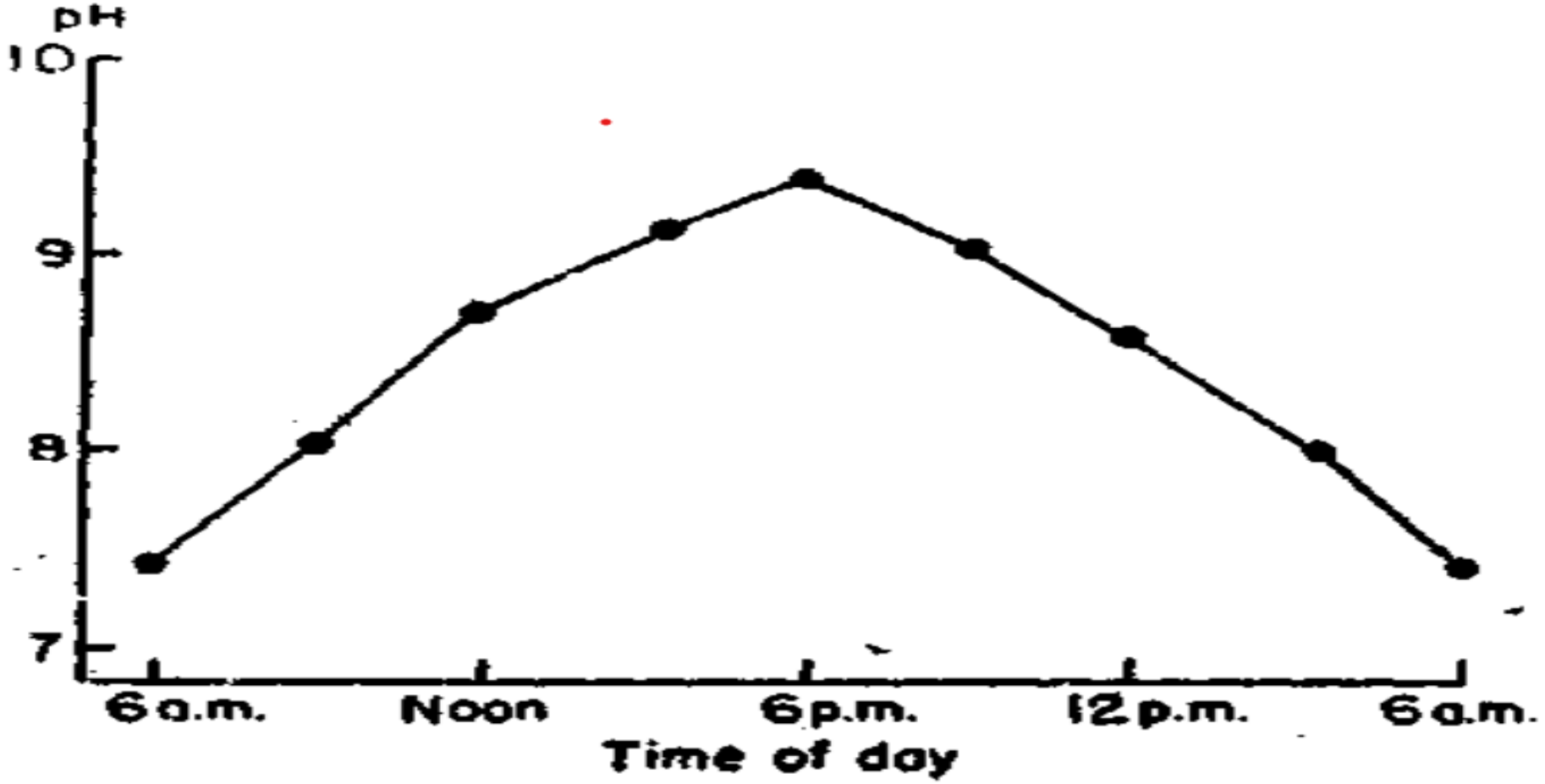


FIG. 1.12. Daily fluctuations in pH in a fish culture pond.

WATER QUALITY MANAGEMENT IN AQUACULTURE

3. Ammonia

- Ammonia reaches pond water as a product of fish metabolism and decomposition of organic matter by bacteria.
- In water, ammonia nitrogen occurs in two forms, un-ionized ammonia and ammonium ion.
- High ammonia levels can arise from overfeeding, protein rich, excess feed decays to liberate toxic ammonia gas, which in conjunction with the fishes, excreted ammonia may accumulate in ponds

WATER QUALITY MANAGEMENT IN AQUACULTURE

3. Ammonia

- As pH rises, un-ionized ammonia (NH_3) increases relative to ammonium ion.
- Temperature also causes an increase in the proportion of un-ionized ammonia, but the effect of increasing temperature is less than that of increasing pH.

WATER QUALITY MANAGEMENT IN AQUACULTURE

3. Ammonia

- As ammonia concentrations increase in the water, ammonia excretion by fish diminishes and levels of ammonia in blood and other tissue increases
- Ammonia increases oxygen consumption by tissues, damages gills and reduces the ability of blood to transport oxygen

WATER QUALITY MANAGEMENT IN AQUACULTURE

3. Ammonia

- Histological changes occur in the gills, kidneys, spleen, thyroid and blood of fish exposed to sublethal concentrations of ammonia
- Disease susceptibility also increases in fish exposed to sublethal concentrations of ammonia

WATER QUALITY MANAGEMENT IN AQUACULTURE

3. Ammonia

- Ammonia concentrations are partially ‘curbed’ or ‘buffered’ by conversion to nontoxic nitrate (No3 -) ion by nitrifying bacteria.
- Additionally, ammonia is converted from toxic ammonia (NH3) to nontoxic ammonium ion (NH4 +)) at pH below 8.0

$$NH_3 + H_2O \quad \Leftrightarrow \quad NH_4^+ + OH^-$$

Low temperature and low pH $\Rightarrow \Leftarrow$ High temperature and high pH

The temperature and pH of water affects the ratio of (NH_4^+):(NH_3) in water.

Simple questions

1. At what time of the day do you expect water pH in ponds to be high? A) 6 a m B) 6 p m C) 12 pm

2. Why do we lime fish ponds A) To provide food to the fish C) To stimulate phytoplankton growth D) To neutralise pH and disinfect the ponds

3. An increase in ammonia concentrations causes the fish to have less ammonia release from the body. **A) True B) False**